

Name : Shan

Sap : 72387

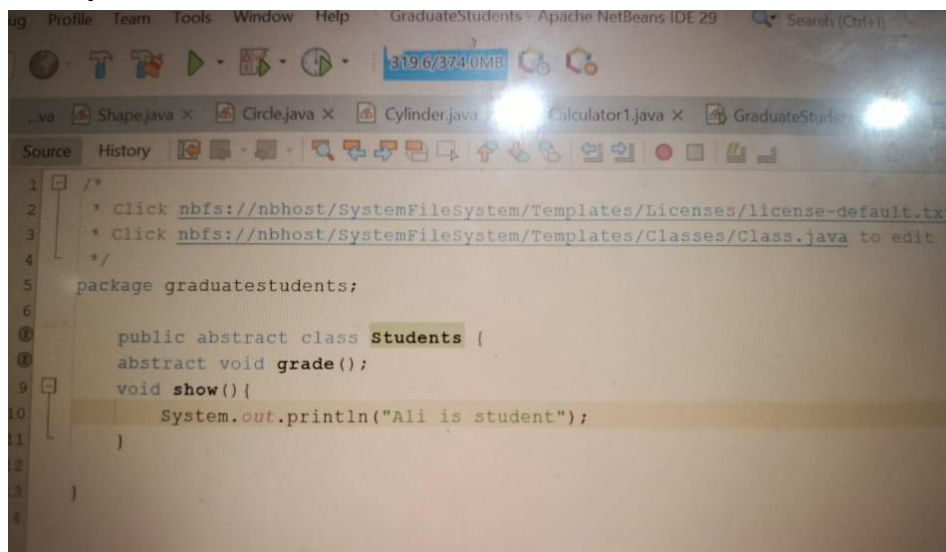
BSAI

OOP (Lab)

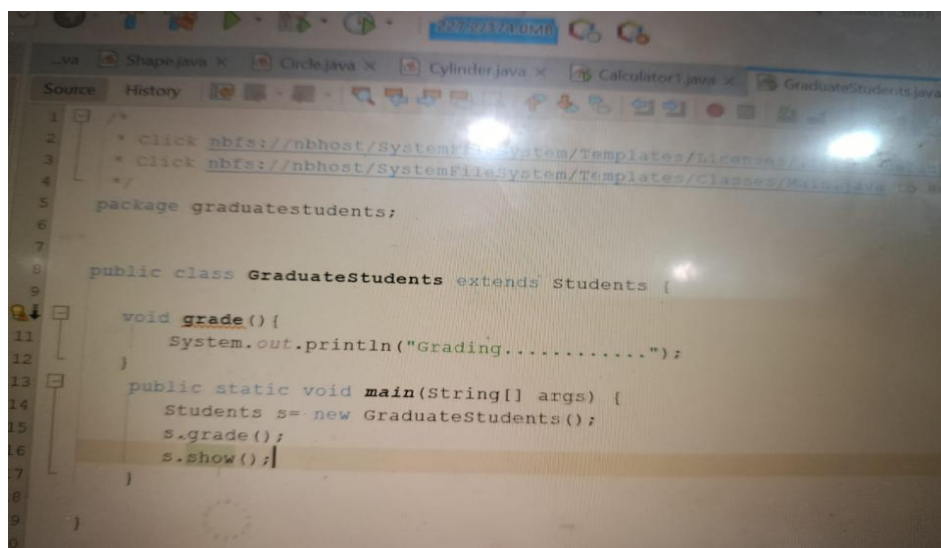
◆ LAB Task 12

✧ Q1: Abstartion Code Simple.

➤ Input

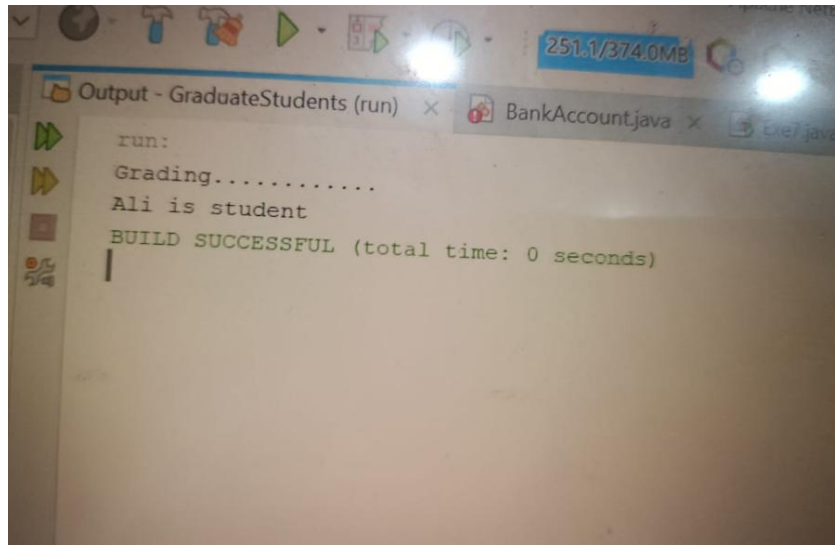


```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit t
4   */
5  package graduatestudents;
6
7  public abstract class Students {
8      abstract void grade();
9      void show() {
10         System.out.println("Ali is student");
11     }
12 }
```



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Main.java to edit
4   */
5  package graduatestudents;
6
7
8  public class GraduateStudents extends Students {
9
10     void grade() {
11         System.out.println("Grading.....");
12     }
13
14     public static void main(String[] args) {
15         Students s= new GraduateStudents();
16         s.grade();
17         s.show();
18     }
19 }
```

➤ Output



- ✧ **Q2:** Write a Java program to create an abstract class **Employee** with abstract methods **calculateSalary()** and **displayInfo()**. Create subclasses **Manager** and **Programmer** that extend the **Employee** class and implement the respective methods to calculate salary and display information for each role.

➤ Input

Employee.java (Abstract Parent)

```
package employeeabstract;

public abstract class Employee {

    String name;

    int id;

    Employee(String name, int id) {

        this.name = name;

        this.id = id;

    }

    abstract double calculateSalary();

    abstract void displayInfo();

}
```

Manager.java (Child 1)

```
package employeeabstract;

public class Manager extends Employee {

    double baseSalary;

    double bonus;

    Manager(String name, int id, double baseSalary, double bonus) {

        super(name, id);

        this.baseSalary = baseSalary;

        this.bonus = bonus;

    }

    double calculateSalary() {

        return baseSalary + bonus;

    }

    void displayInfo() {

        System.out.println("--- Manager Info ---");

        System.out.println("Name   : " + name);

        System.out.println("ID    : " + id);

        System.out.println("Salary : " + calculateSalary());

    }

}
```

Programmer.java (Child 2)

```
package employeeabstract;

public class Programmer extends Employee {

    double hourlyRate;

    int hoursWorked;

    Programmer(String name, int id, double hourlyRate, int
hoursWorked) {

        super(name, id);

        this.hourlyRate = hourlyRate;

        this.hoursWorked = hoursWorked;

    }

}
```

```

    }

    double calculateSalary() {

        return hourlyRate * hoursWorked;

    }

    void displayInfo() {

        System.out.println("--- Programmer Info ---");

        System.out.println("Name   : " + name);

        System.out.println("ID    : " + id);

        System.out.println("Salary : " + calculateSalary());

    } }

```

Main.java

```

package employeeabstract;

public class Main {

    public static void main(String[] args) {

        Employee m = new Manager("Shan", 72387, 150000, 30000);

        m.displayInfo();

        System.out.println();

        Employee p = new Programmer("Ali", 10001, 500, 200);

        p.displayInfo();

    } }

```

➤ Output

```

--- Manager Info ---
Name       : Shan
ID         : 72387
Salary     : 180000.0

--- Programmer Info ---
Name       : Ali
ID         : 10001
Salary     : 100000.0

```

- ✧ **Q3:** Write a Java program to create an abstract class **BankAccount** with abstract methods **deposit()** and **withdraw()**. Create subclasses: **SavingsAccount** and **CurrentAccount** that extend the **BankAccount** class and implement the respective methods to handle deposits and withdrawals for each account type.

➤ **Input**

BankAccount.java (Abstract Parent)

```
package bankaccount;

public abstract class BankAccount {

    String accountHolder;

    int accountNumber;

    double balance;

    BankAccount(String accountHolder, int accountNumber, double
balance) {

        this.accountHolder = accountHolder;

        this.accountNumber = accountNumber;

        this.balance = balance;

    }

    abstract void deposit(double amount);

    abstract void withdraw(double amount);

    void showBalance() {

        System.out.println("Current Balance : " + balance);

    } }

}
```

SavingsAccount.java (Child 1)

```
package bankaccount;

public class SavingsAccount extends BankAccount {

    double minimumBalance;

    SavingsAccount(String accountHolder, int accountNumber,

        double balance, double minimumBalance) {
```

```

        super(accountHolder, accountNumber, balance);

        this.minimumBalance = minimumBalance;
    }

    void deposit(double amount) {

        balance = balance + amount;

        System.out.println("Savings Deposit : +" + amount);

        showBalance();
    }

    void withdraw(double amount) {

        if (balance - amount < minimumBalance) {

            System.out.println("Withdrawal FAILED! Minimum balance "

                + minimumBalance + " rehna zaroori hai.");

        } else {

            balance = balance - amount;

            System.out.println("Savings Withdrawal : -" + amount);

            showBalance();

        }

    }

    void displayInfo() {

        System.out.println("===== Savings Account =====");

        System.out.println("Name      : " + accountHolder);

        System.out.println("Account No   : " + accountNumber);

        System.out.println("Minimum Balance: " + minimumBalance);

        showBalance();

    } }

```

CurrentAccount.java (Child 2)

```
package bankaccount;
```

```

public class CurrentAccount extends BankAccount {

    double overdraftLimit;

    CurrentAccount(String accountHolder, int accountNumber,
                    double balance, double overdraftLimit) {

        super(accountHolder, accountNumber, balance);

        this.overdraftLimit = overdraftLimit;
    }

    void deposit(double amount) {

        balance = balance + amount;

        System.out.println("Current Deposit : +" + amount);

        showBalance();
    }

    void withdraw(double amount) {

        if (balance - amount < -overdraftLimit) {

            System.out.println("Withdrawal FAILED! Overdraft limit "
                               + overdraftLimit + " cross ho jayegi.");

        } else {

            balance = balance - amount;

            System.out.println("Current Withdrawal : -" + amount);

            showBalance();
        }
    }

    void displayInfo() {

        System.out.println("===== Current Account =====");

        System.out.println("Name      : " + accountHolder);

        System.out.println("Account No   : " + accountNumber);

        System.out.println("Overdraft Limit: " + overdraftLimit);

        showBalance();
    }
}

```

Main.java

```
package bankaccount;

public class Main {

    public static void main(String[] args) {

        SavingsAccount sa = new SavingsAccount("Shan", 72387,
50000, 10000);

        sa.displayInfo();

        System.out.println();

        sa.deposit(5000);

        sa.withdraw(10000);

        sa.withdraw(40000);

        System.out.println();

        CurrentAccount ca = new CurrentAccount("Global Link
Traders", 10001, 20000, 15000);

        ca.displayInfo();

        System.out.println();

        ca.deposit(8000);

        ca.withdraw(30000);

        ca.withdraw(20000);

    } }
```


➤ Output

```
===== Savings Account =====
Name           : Shan
Account No     : 72387
Minimum Balance: 10000.0
Current Balance : 50000.0

Savings Deposit : +5000.0
Current Balance : 55000.0
Savings Withdrawal : -10000.0
Current Balance : 45000.0
Withdrawal FAILED! Minimum balance 10000.0 rehna zaroori hai.

===== Current Account =====
Name           : Global Link Traders
Account No     : 10001
Overdraft Limit: 15000.0
Current Balance : 20000.0

Current Deposit : +8000.0
Current Balance : 28000.0
Current Withdrawal : -30000.0
Current Balance : -2000.0
Withdrawal FAILED! Overdraft limit 15000.0 cross ho jayegi.
```